Assembly Manual

Licence plate detection with Jetson Nano

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Acronyms

1 General Assembly Hints

Before beginning the assembly of the License Plate Detection System, please read the following important guidelines. These instructions ensure a safe, efficient, and error-free setup process for the Jetson Nano hardware and its connected components.

Preparation and Safety Measures

- **Work Environment:** Assemble all components on a clean, dry, and non-conductive surface. Ensure proper lighting and ventilation to avoid overheating during operation.
- **Electrostatic Discharge (ESD):** Handle the Jetson Nano and camera only by their edges. Use an antistatic wrist strap or touch a grounded metal object before handling any electronics.
- **Cable Management:** Arrange all cables neatly and ensure that they are not under tension. Avoid crossing power and data cables where possible.
- **Component Orientation:** Check the correct direction of all connectors (micro-SD, USB, HDMI, and power) before insertion. Do not apply excessive force.
- **Power Supply:** Use only the official 5 V - 4 A adapter. Always disconnect power before adding or removing components.

Table 1.1: Assembly Preparation Checklist

Item	**Description / Action Required**
Work Surface	Clean, dry, non-conductive, and well-lit area
Power Adapter	Verified 5V - 4A supply (official Jetson Nano adapter)
SD Card	32 GB card pre-flashed with JetPack 5.1.2 (Ubuntu 20.04)
Display and Peripherals	HDMI monitor, USB keyboard, and mouse ready
Camera Module	Microsoft LifeCam Show (USB) connected but protected with lens cover
Network Connection	Ethernet or Wi-Fi adapter available for first boot
Static Safety	ESD wrist strap or grounding point nearby

Ventilation and Safety

- Place the Jetson Nano on a hard, flat surface with sufficient airflow around the heat sink.
- Do not operate the system inside a closed enclosure without ventilation holes.
- Avoid touching the heat sink during or immediately after operation, as it may become hot.
- If long-duration tests are performed, use an external cooling fan for optimal performance.

Post-Assembly Verification

After completing all connections, verify the following points:

1. The power LED on the Jetson Nano turns on when the adapter is connected.
2. The HDMI display output is visible on the monitor.
3. The USB camera is detected and functioning properly.
4. All cables and connectors are securely attached.
5. No abnormal heat or noise occurs during operation.

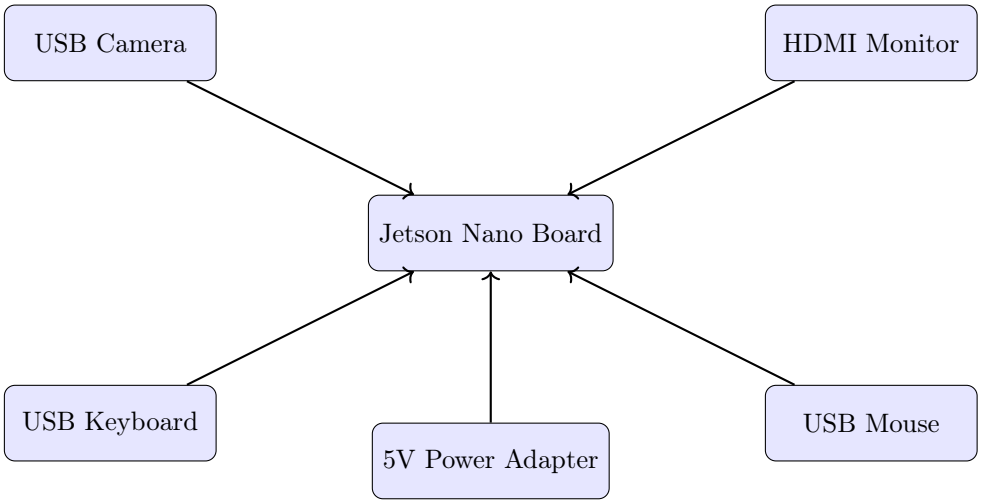

Figure: Assembly layout for Jetson Nano hardware setup.

Following these general assembly hints ensures reliable installation, minimizes hardware risks, and provides a stable foundation for further software configuration and testing.

2 Main Function

The assembled unit implements real-time license plate acquisition on the NVIDIA Jetson Nano. The USB camera captures frames, the Jetson performs detection and optical character recognition, and results can be visualized on the HDMI display. This chapter describes the overall function, physical interconnects, and power path.

Overview

- **Compute Core:** Jetson Nano (4 GB) processes video frames locally.
- **Image Acquisition:** Microsoft LifeCam Show (USB) delivers the input stream.
- **User I/O:** HDMI monitor for display; USB keyboard and mouse for setup.
- **Storage:** 32 GB microSD with JetPack 5.1.2.
- **Power:** 5 V 4 A DC adapter.

Table 2.1: Interfaces Used by the Assembly

Subsystem	**Jetson Port**	**Notes**
USB Camera	USB 2.0/3.0 Type-A	Connect directly to a USB port on Jetson Nano.
Display	HDMI	Use a short high-quality HDMI cable.
User Input	USB Type-A	Keyboard and mouse on separate ports or via USB hub.
Network (optional)	Ethernet (RJ45)	Recommended for first setup and package updates.
Power	DC Barrel Jack	Official 5 V 4 A adapter (5 V DC).
Storage	microSD slot	32 GB microSD with JetPack 5.1.2.

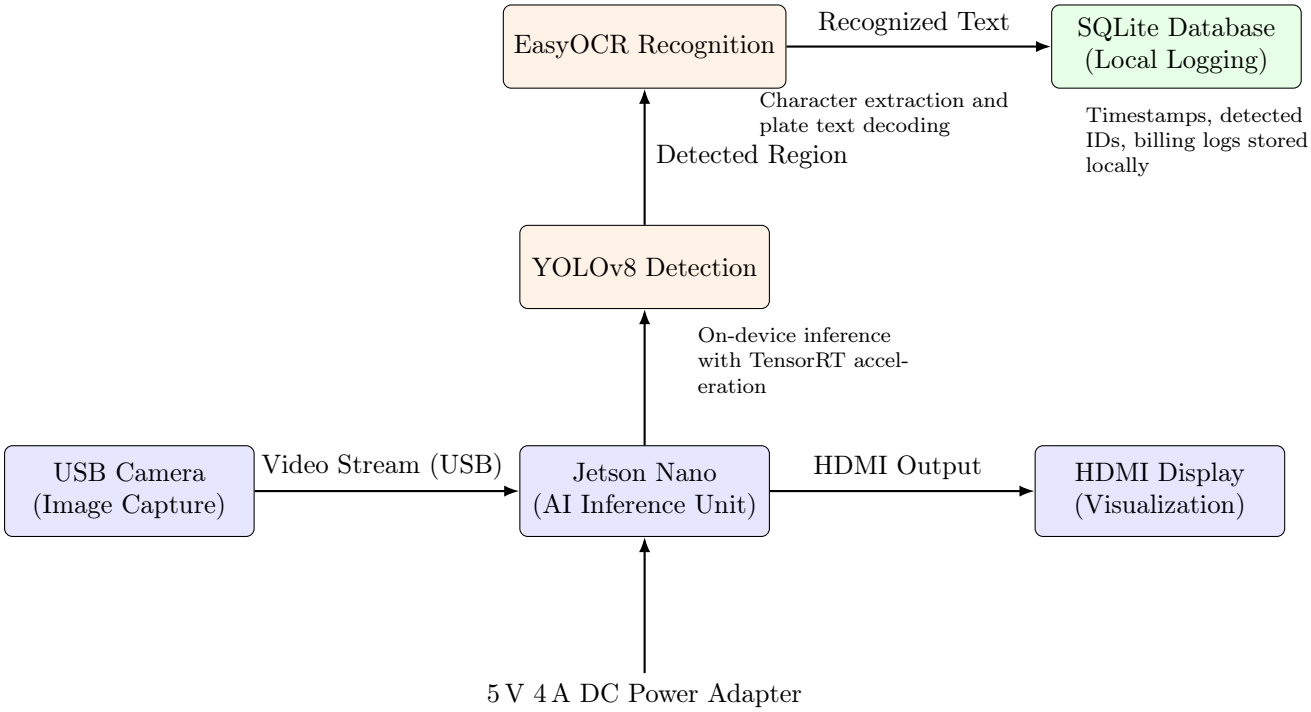

Figure 2.1: Overall license plate detection workflow implemented on Jetson Nano: from image capture to OCR and local database logging.

This diagram visually represents the complete end-to-end workflow of the system and the logical interaction between its key modules. It begins with the **USB camera**, which continuously captures live video frames from the environment. These frames are transmitted through the USB interface to the **Jetson Nano**, which serves as the system's central AI processing unit. Inside the Nano, the incoming image stream is processed by a pre-trained **YOLOv8 model** optimized for edge inference using TensorRT acceleration. Once a vehicle license plate is detected, the corresponding image region is forwarded to the **EasyOCR recognition layer**, which extracts alphanumeric text with high precision even under varying lighting conditions. The recognized plate numbers, along with timestamps, detection confidence scores, and event details, are then stored in the local **SQLite database** for persistent on-device logging and later retrieval. The output from the Jetson Nano is displayed on an **HDMI monitor** for live visualization of detection results, providing both verification and debugging feedback for the operator. The entire pipeline is powered by a regulated 5 V, 4 A DC adapter that ensures stable GPU performance, thermal efficiency, and uninterrupted operation during continuous video inference. This workflow clearly outlines the system's real-time nature—combining data acquisition, intelligent detection, optical recognition, and data management—all within a compact and power-efficient embedded platform.

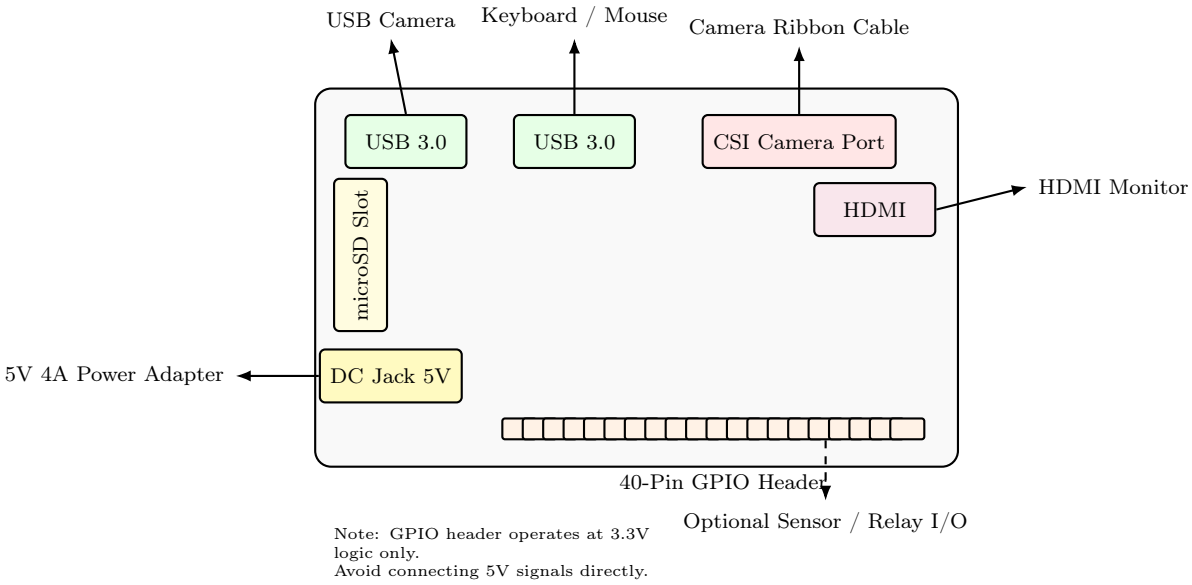

Figure 2.2: Top-view Jetson Nano hardware layout showing only the active connections used in the license plate detection system — USB camera, HDMI display, keyboard/mouse, CSI camera, GPIO, microSD, and 5V DC power.

The above top-view layout illustrates the exact hardware configuration of the assembled system. It shows how the USB camera, HDMI monitor, keyboard, and mouse are connected to the Jetson Nano through their respective ports, while power is supplied via a 5 V DC adapter. The CSI camera interface and GPIO header are highlighted for potential future extensions, and the microSD card slot provides storage for the operating system and project files. Arrows indicate the direction of power and data flow, ensuring that the physical connections match the functional workflow shown in the previous diagram.

Table 2.2: Cable and Connector Checklist

Connection	**Checkpoints**
USB Camera → Jetson	Firm seating, no strain; short cable preferred.
HDMI → Monitor	Cable fully inserted; monitor set to correct input source.
Keyboard / Mouse → Jetson	Separate ports or powered hub; verify LEDs (if present).
Ethernet → Router/Switch	Link LEDs active; stable connection for first boot.
DC Adapter → Jetson	Official 5 V 4 A; avoid loose barrel jack.

3 Sketch

4 List Of Parts

5 Set Up

relocation and quick system reassembly at different deployment sites.

6 Functions

7 Maintenance

8 Safety Guidelines

9 Troubleshooting

10 Add Ons

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